
AR 507: Probabilistic Robotics

Course Code	AR 507
Course Name	Probabilistic Robotics
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Probability Theory and Linear Algebra:** Sample space and events, Conditional probability, Expected value and variance. Uniform, normal, exponential random variables. Systems of linear equations, Linear dependence and independence, Operations with Matrices, Eigenvalues and eigenvectors. (6 hours)
- **Robot Motion:** Probabilistic kinematics, Velocity motion model, Odometry motion model. (4 hours)
- **Sensors for robotics:** Coordinate frame transformations, camera model, camera calibration, Sonar, Lidar, GPS, etc. (5 hours)
- **Recursive State Estimation:** Bayesian filter, Kalman filter (KF), EKF, & Particle filter. (11 hours)
- **Robot Localization, Mapping, and SLAM:** Localization problems, Markov localization, EKF localization, Grid localization, Monte Carlo localization, Occupancy grid mapping algorithm, EKF SLAM. (16 hours)

Text Books

1. Sebastian Thrun, Wolfram Burgard and Dieter Fox, **Probabilistic Robotics**, MIT press, 2005.
2. Papoulis A. and Pillai S. U., Probability, **Random Variable, and Stochastic Processes**.

References

1. **Probabilistic Robotics:** <http://www.probabilistic-robotics.org/>
2. Strang G., **Linear Algebra and its Applications**.
3. Ron Larson, **Calculus: Elementary Linear Algebra**, 8th Edition, Cengage Learning, 2017.